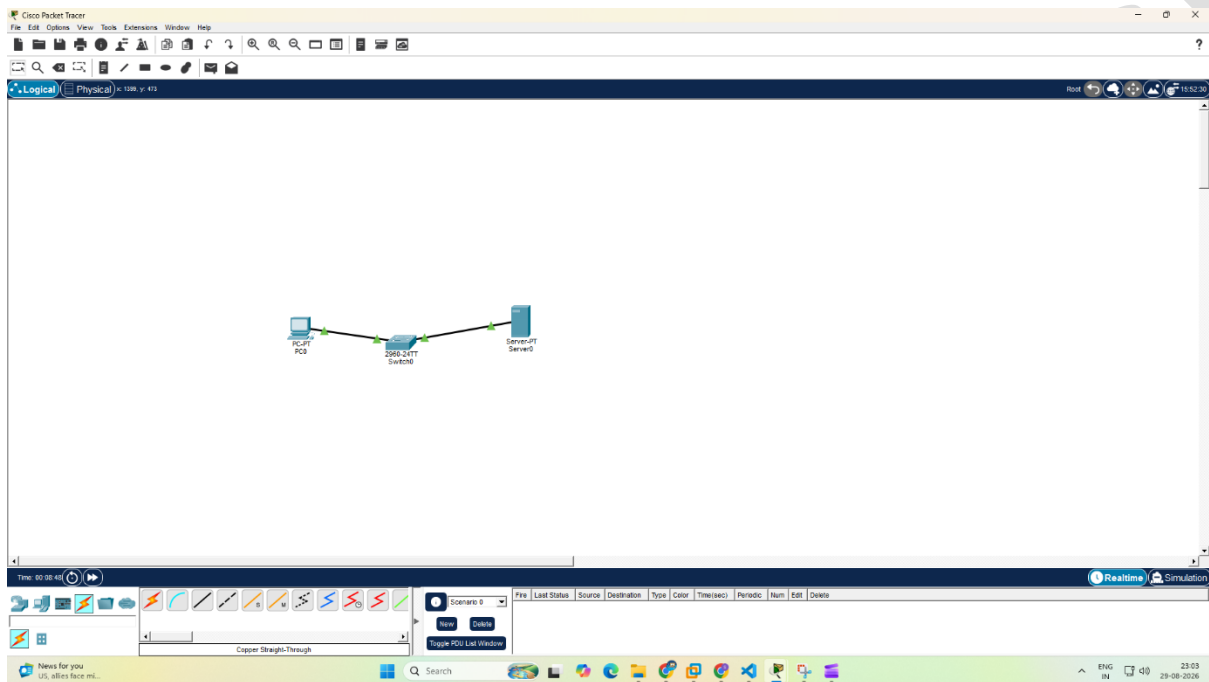


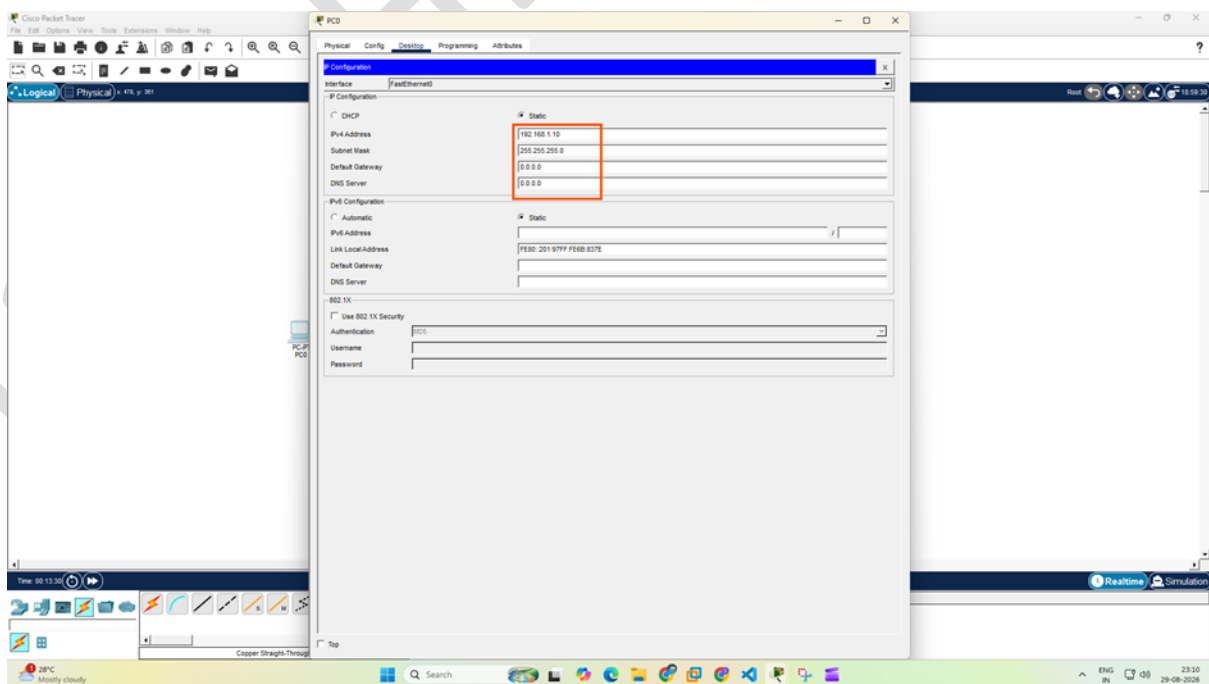
## Practical 7

**Aim :** Configure and implement secure web communication using SSL/TLS protocols, including certificate management and secure session establishment.

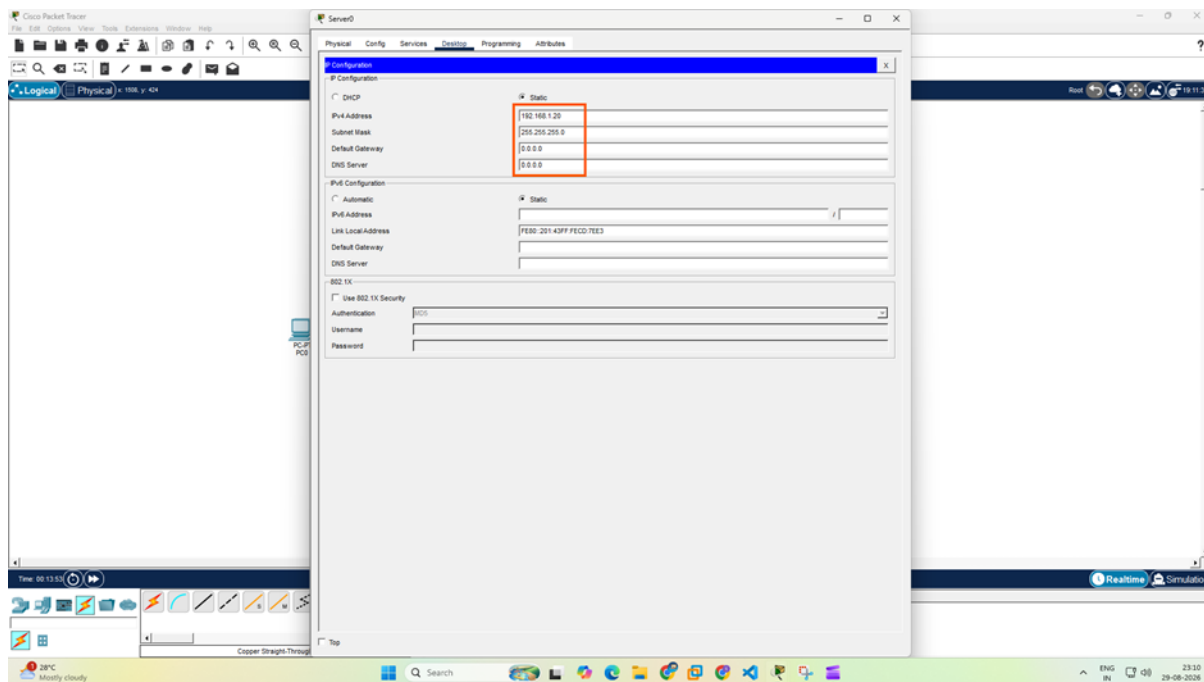
**Step 1:** Open Cisco Packet Tracer and add **PC0**, **Server0**, and a **2960 Switch** to the workspace. Connect **PC0** and **Server0** to **Switch0** using **Copper Straight-Through** cables, then wait for the connection indicators to turn green.



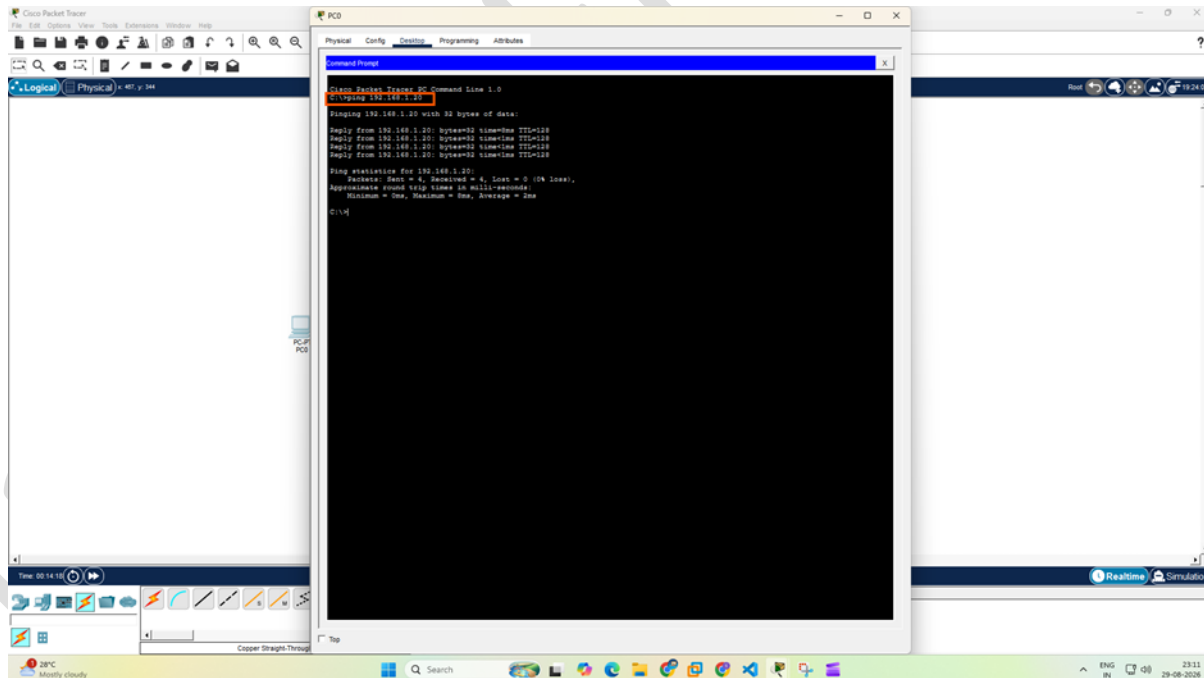
**Step 2:** Go to **PC0** → **Desktop** → **IP Configuration** and set IP Address to **192.168.1.10** and Subnet Mask to **255.255.255.0**. Leave the **Default Gateway** blank because both devices are on the same network.



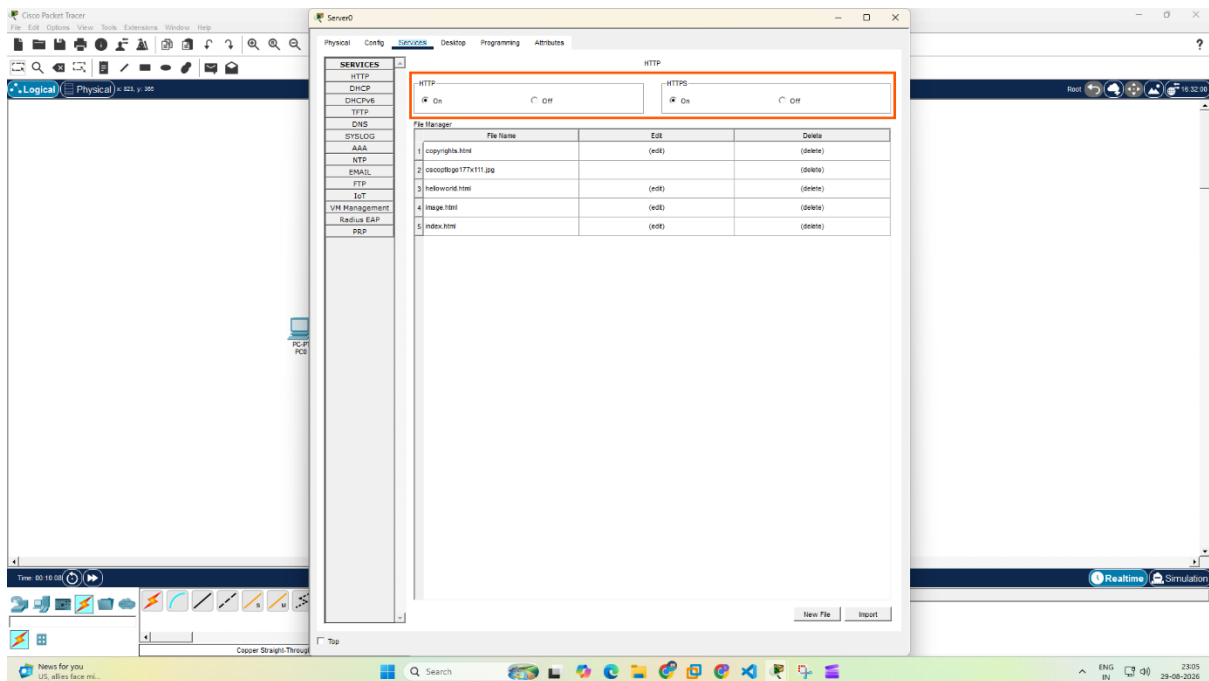
**Step 3:** Go to **Server0** → **Desktop** → **IP Configuration** and set IP Address to **192.168.1.20** and Subnet Mask to **255.255.255.0**.  
Leave the **Default Gateway** blank. Network: **192.168.1.0/24**.



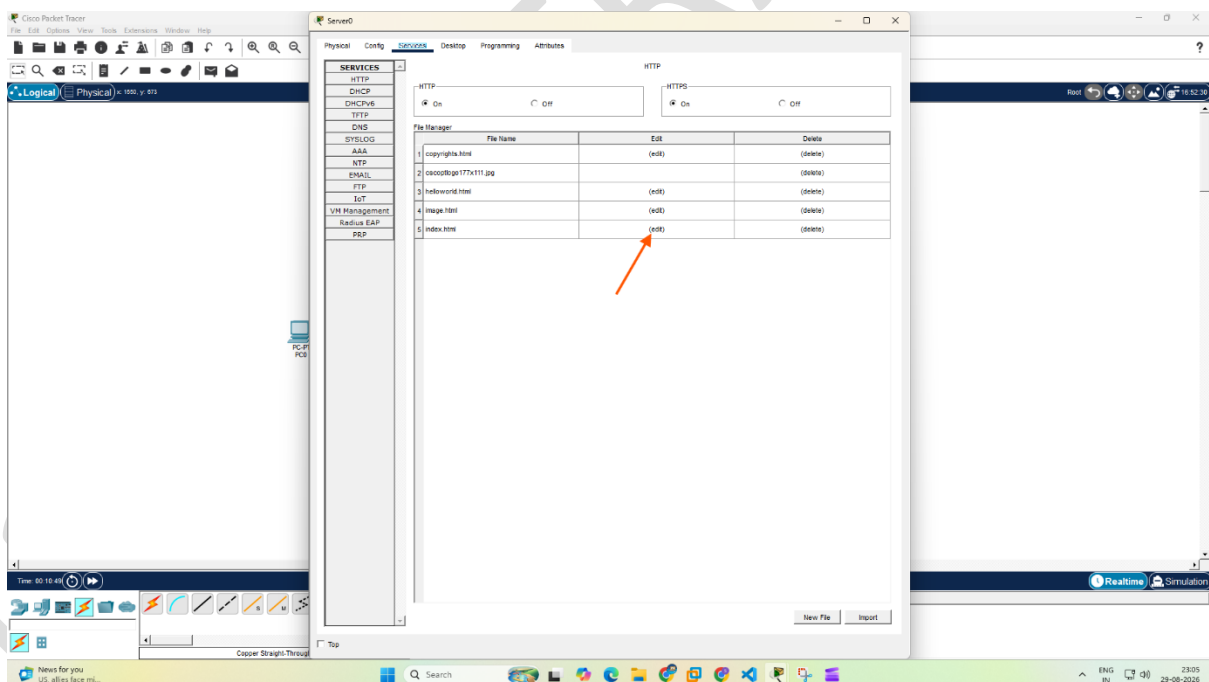
**Step 4:** Go to **PC0** → **Desktop** → **Command Prompt** and type ping 192.168.1.20.  
If you get **"Reply from 192.168.1.20"**, the PC and Server are successfully communicating.

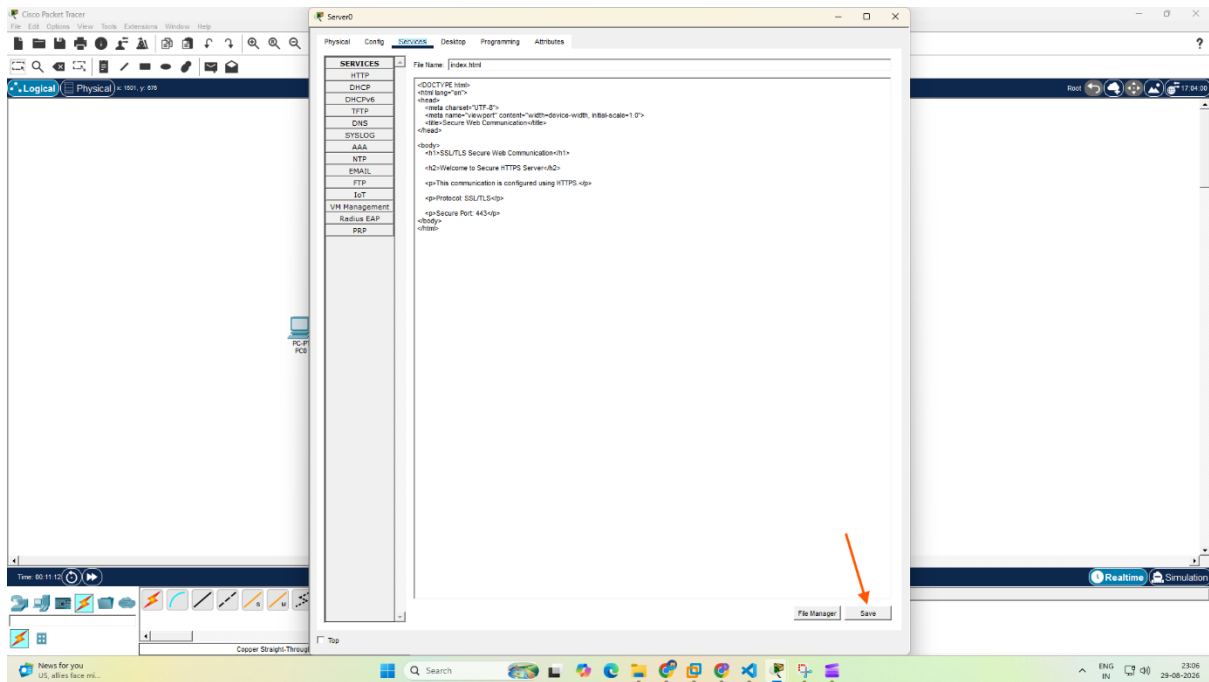


**Step 5:** Go to **Server0** → **Services** → **HTTP** and turn **HTTP: ON** and **HTTPS: ON**.  
The server will now support **HTTP (Port 80)** and **HTTPS (Port 443)**.

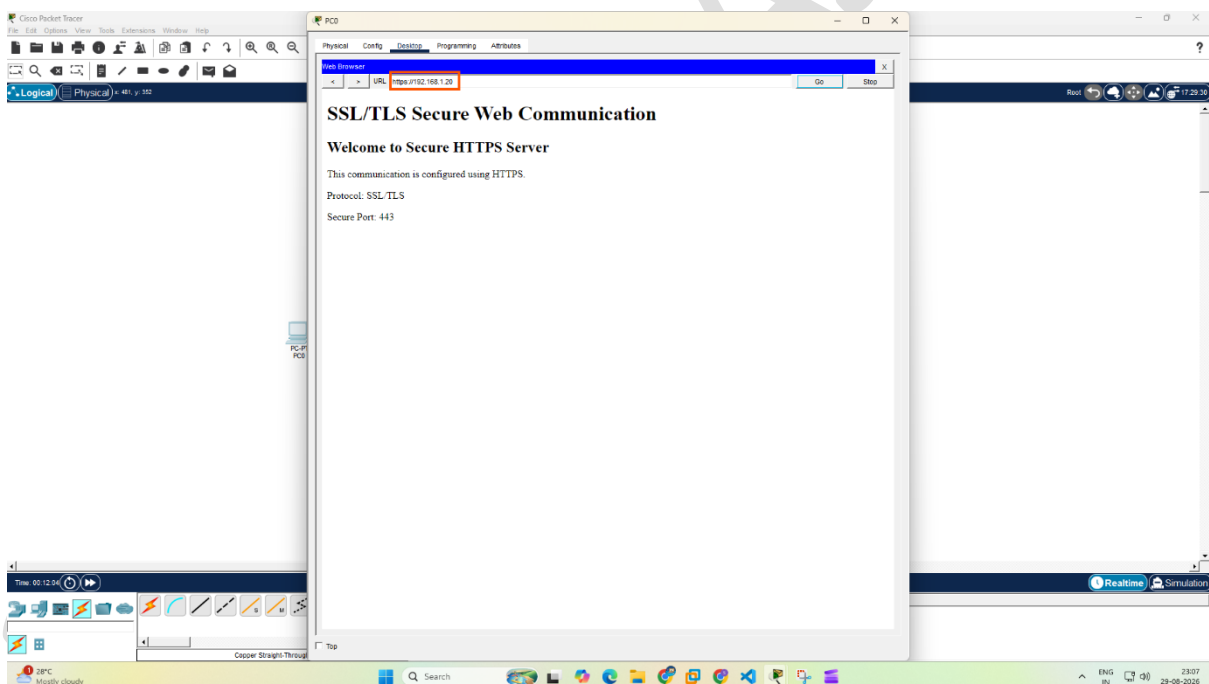


**Step 6:** Go to **Server0** → **Services** → **HTTP** and edit the HTML web page as needed.  
Click **Save** to apply the changes.

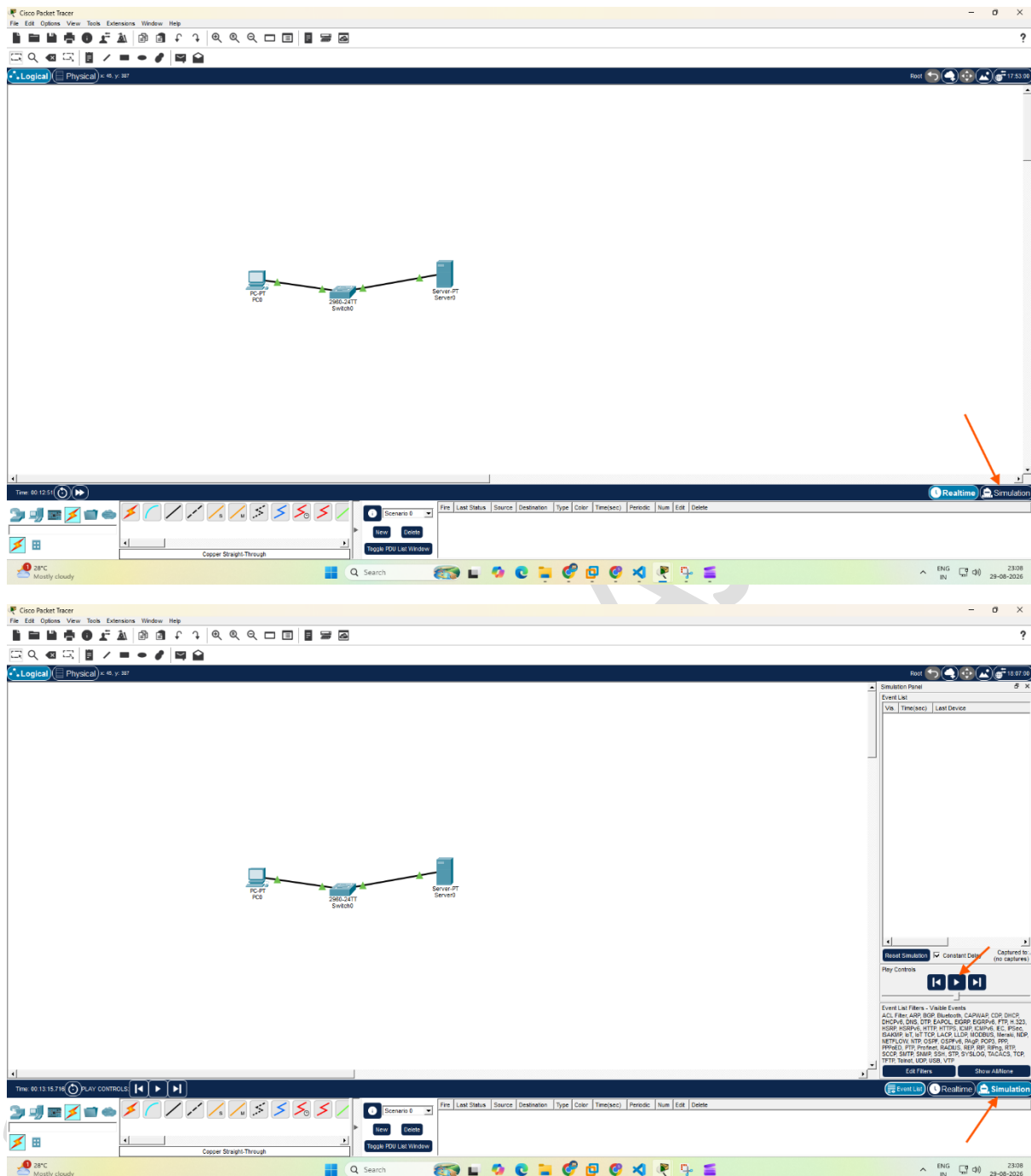




**Step 7:** Go to PC0 → Desktop → Web Browser and enter <https://192.168.1.20>.  
If the web page opens, HTTP communication is working on **Port 443**.



**Step 8:** In Cisco Packet Tracer, switch from **Realtime** to **Simulation** mode using the bottom-right option. This allows you to **observe network communication step-by-step**.



**Step 9:** Enter <https://192.168.1.20> in **PC0** → **Web Browser** and observe the communication through **PC0** → **Switch0** → **Server0**.

The server provides secure web service using **HTTPS (Port 443)**.

